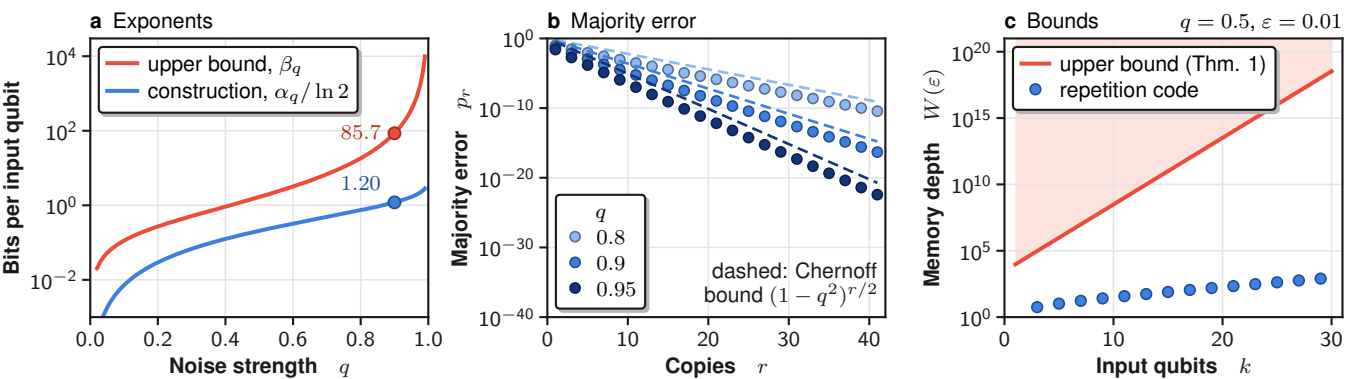

a Exponents
upper bound, β_q
construction, $\alpha_q / \ln 2$
85.7
1.20
Bits per input qubit
Noise strength q
b Majority error
Majority error p_r
q
0.8
0.9
0.95
dashed: Chernoff bound $(1-q^2)^{r/2}$
Copies r
c Bounds
$q = 0.5, \varepsilon = 0.01$
upper bound (Thm. 1)
repetition code
Memory depth $W(\varepsilon)$
Input qubits k